

Jie Fu 傅杰

Ph.D. Student in Mathematics, Qiuzhen College, Tsinghua University
fu-j21@mails.tsinghua.edu.cn | fujiemath.com | github.com/vegetablefj
Tsinghua University, Beijing, China

PERSONAL STATEMENT

I am Jie Fu (傅杰 in Chinese). I was born in March 2006. I am now a PhD student at Qiuzhen College, Tsinghua University.

RESEARCH INTERESTS

Algebraic geometry and related areas, with current work on (computations of) automorphism groups of hypersurfaces and hyperkähler manifolds. Further interests include moduli theory, rationality problems, quantum cohomology, Bridgeland stability, and mirror symmetry.

EDUCATION

2025–present	Tsinghua University , Qiuzhen College, Beijing, China. Ph.D. student in Mathematics. Advisor: Prof. Zhiwei Zheng.
2021–2025	Tsinghua University , Qiuzhen College, Beijing, China. Bachelor in Mathematics and Applied Mathematics.

PUBLICATIONS AND PREPRINTS

Preprints

- [1] **Non-symplectic Indices of Automorphism Groups of Smooth Cubic Fourfolds.**
With Shihao Wang and Zhiwei Zheng. *arXiv preprint*, arXiv:2606.11754 [math.AG], 2026. 30 pages.
arxiv.org/abs/2606.11754

TALKS, SEMINARS AND ACADEMIC ACTIVITIES

- *Smooth hypersurface with automorphism group of large order.* Spring Learning Seminar in Algebraic Geometry, YMSC, Beijing. March 18th, 2026.
- *New results on automorphism groups of cubic fourfolds.* YMSC Seminar on Moduli Spaces and Related Topics, YMSC, Beijing. March 12th, 2026.
- *Symplectic automorphism of $K3^{[n]}$ -type manifolds.* Hyperkähler Seminar, YMSC, Beijing. December 2nd, 2025.
- *Linear system of $K3$ surfaces.* Hyperkähler Seminar, YMSC, Beijing. October 24th, 2025.
- *Basic lattice theory following Nikulin.* Hyperkähler Seminar, YMSC, Beijing. September 30th, 2025.
- Co-organization: Student Seminar on Perverse Sheaves and Application to Representation Theory, Qiuzhen College, Beijing, Fall 2025.

TEACHING EXPERIENCE

Spring 2026 Teaching assistant, *Algebraic Variety* given by Prof. Junpeng Jiao, Tsinghua University.

HONORS AND AWARDS

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| Spring 2026 | Pi day's best problems award, Qiuzhen College. |
| Spring 2023 | Honorable Mention, Analysis and Differential Equations, S.-T. Yau College Student Mathematics Contest. |
| Fall 2020 | First prize in China's national high school mathematics competition, Chinese Mathematical Society. |

TECHNICAL SKILLS

Computational Software	GAP; SageMath; Magma; OSCAR/Julia.
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REFERENCES

Prof. Zhiwei Zheng	Advisor. Asistant Profosser, Tsinghua University, Yau Mathematical Sciences Center, zhengzhiwei@mail.tsinghua.edu.cn
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